higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

COMMUNICATIONS ELECTRONICS N5

(8080235)

15 November 2019 (X-Paper)
09:00–12:00

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
COMMUNIVATION ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1:

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–N) next to the question number (1.1–1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	The only component available to oppose current flow in a direct current circuit	A	folded dipole
1.2	A component that sets up a back emf when AC flows through it, which is opposite to the applied emf	B	L-type
1.3	A two-port network which has similar input and output ports that may be interchanged	C	Shannon Hartley
1.4	The network pads used for matching purposes only and are designed to give minimum attenuation	D	resistance
1.5	A network that is designed to pass a certain band of frequencies without loss, while attenuating all other frequencies	E	symmetrical
1.6	A stage in an AM transmitter which generates the constant amplitude which will be modulated and transmitted	F	coil
1.7	The modulation process that produces two set of side bands and is a narrow- band system	G	AM
1.8	The theorem that specifies the limit of the rate of speed or channel capacity, that data may be transmitted for a given bandwidth within a certain signal-to-noise ratio	H	oscillator
1.9	The ratio of power radiated by the antenna to the square of the current at the feed point	I	radiation resistance
1.10	The antenna that offers high resistance to the circuit or transmission line	J	half-wave dipole
		K	filter
		L	T-type
		M	reactance
		N	FM

(10 × 1) **[10]**

QUESTION 2: AC NETWORKS

2.1 Draw the circuit diagrams of a parallel resonance circuit which is connected to a 100 V alternating current supply with a series resistance of 5 Ω and inductor of 585 µH connected in parallel to a capacitor of 120 pF.

Determine the following:

2.1.1 Drawing of circuit (2)

2.1.2 The dynamic impedance (2)

2.1.3 The Q factor of the circuit (2)

2.1.4 The current flowing through the capacitor (4)

2.1.5 The frequency at resonance (2)

2.2 Use an alternate formula to find the Q factor in QUESTION 2.1.3 (4)

2.3 Define *co-efficient of coupling*. (4)

2.4 Consider the rules of interaction for mutually inductive coupled circuits and fill in the missing words by choosing the words from the given list below to complete the sentences in question number (2.4.1(a)–2.4.3(c)) in the ANSWER BOOK.

induced; current; coupled; primary; lags; impedance; reflected; winding; voltage; secondary.

2.4.1 As far as the (a) ... circuit is concerned the effect of the secondary circuit is as if an/a (b) ... has been added in series with the primary impedance and is called the (c) … or (d) … impedance. (4)

2.4.2 The (a) ... induced in a secondary circuit by the primary (b) ... as a magnitude of WMIp and (c) ... behind the current by 90 °C. (3)

2.4.3 The (a) ... current is exactly the same current that will flow if the (b) ... voltage was applied in series with the secondary circuit and the primary (c) ... was absent. (3)

[30]

QUESTION 3: FOUR TERMINAL NETWORKS

3.1 Find the characteristic impedance of the network in FIGURE 1 below. (6)

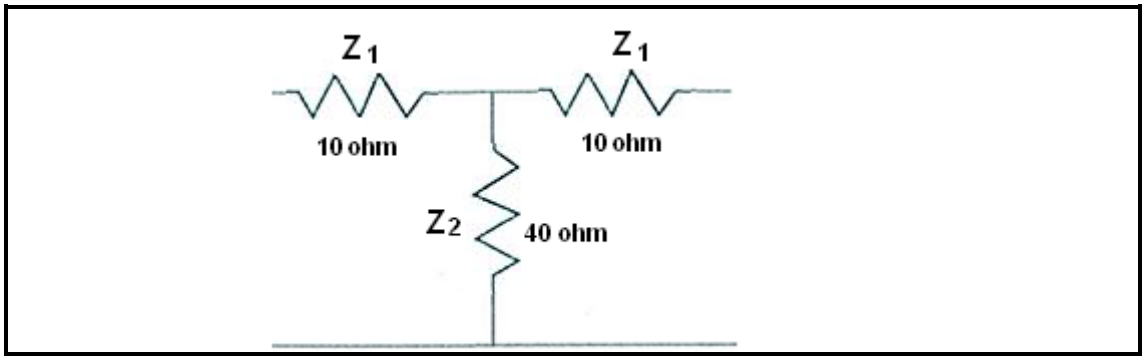


FIGURE 1

3.2 An attenuator input is 3 watts and the output is 5 mW.

Calculate the attenuation of the pad in nepers. (4)

[10]

QUESTION 4: FILTERS

4.1 Determine the following:

4.1.1 Identify the filters on FIGURE 1 and FIGURE 2 below. (2)

4.1.2 Calculate the cut- off frequencies for both filters in QUESTION 4.1.1 (3)

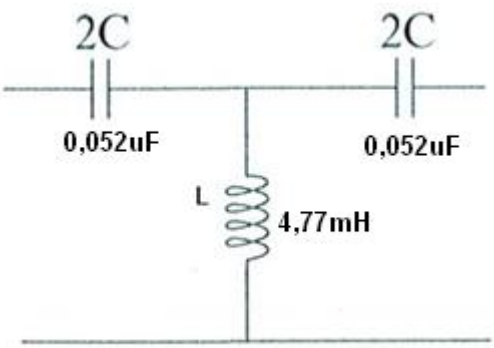


FIGURE 1

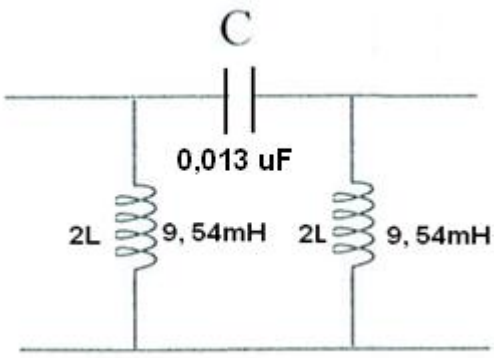


FIGURE 2

4.2 Design a T-type low pass filter having a cut-off frequency of 4 kHz for a 600 Ω transmission line. (5)

[10]

QUESTION 5: RECEIVERS AND TRANSMITTERS

5.1 List THREE advantages of double sideband transmission. (3 × 2) (6)

5.2 Draw the neatly labelled block diagram of an AM Superheterodyne receiver. (5)

5.3 Explain the function of each stage of the receiver in QUESTION 5.2. (18)
[29]

QUESTION 6: ANTENNAE

6.1 Give THREE reasons why the ratio of escaping to returning waves is small in antennae. (3)

6.2 Define *effective aperture* of an antenna. (2)
[5]

QUESTION 7: BODE PLOTS

7.1 Define *bode diagrams.* (3)

7.2 Give THREE reasons why phase lead and lag are used in bode plots. (3)
[6]

TOTAL 100

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FORMULA SHEET

Inductive reactance

$X_L = 2\pi f L$

Capacitive Reactance

$X_C = \frac{1}{2\pi f C}$

Series Formulae

$Z_T = \sqrt{(R^2 + (X_L - C)2}$

$\tan\Theta = \frac{X_L - X_C}{R}$

$BW = \frac{f_o}{Q} = \frac{R}{2\pi L}$

$p.f = Cos\ \phi$

$P = VIC\ Cos\ \phi$

$Y = \frac{1}{Z} = G \pm jB$

Parallel Formulae

$I_T = \sqrt{(I^2R + (I_C - I_L)^2}$

$\tan\Theta = \frac{I_C - I_L}{IR}$

$I_X = I_C - I_L$

$p.f = Cos\ \phi$

$P = VIC\ Cos\ \phi$

$Y_T = Y_1 + Y_2$

Series resonance

$f_r = \frac{1}{Z} = G \pm jB$

$Z = R$

$\varphi = 0°$

p.f = Cos ϕ =1

$Q = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$

$BW = \frac{f_o}{Q} = \frac{R}{2\pi L}$

Parallel resonance

$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

$Z_D = \frac{L}{RC}$

$\varphi = 0°$

p.f = Cos ϕ =1

$Q = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$

$BW = \frac{f_o}{Q} = \frac{R}{2\pi L}$

FILTERS

Low pass

$R_o = \sqrt{\frac{L}{C}}$

$f_c = \frac{1}{\pi\sqrt{LC}}$

$C = \frac{1}{\pi R_o f_c}$

$L = \frac{R_o}{\pi f_C}$

$Z_{OT} = \sqrt{\frac{L}{C}}\ X\ \sqrt{1 - \frac{\omega^2 LC}{4}}$

$Z_{O\pi} = \frac{R_o}{\sqrt{1 - \frac{\omega^2 LC}{4}}}$

High pass

$R_o = \sqrt{\frac{L}{C}}$

$f_c = \frac{1}{4\pi\sqrt{LC}}$

$C = \frac{1}{4\pi R_o f_c}$

$L = \frac{R_o}{4\pi F}$

$Z_{OT} = \sqrt{\frac{L}{C}}\ X\ \sqrt{1 - \frac{1}{4\omega^2 LC}}$

$Z_{O\pi} = \frac{R_o}{\sqrt{1 - \frac{1}{4\omega^2 LC}}}$

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SYMMETRICAL FOUR – TERMINAL NETWORK

T – Network

$Z_{in} = Z_{out/uit}$

$Z_{OT} = \sqrt{Z_{oc}Z_{sc}}$

$Z_{OT} = \sqrt{\frac{Z_1^2}{4} + Z_1Z_2} = \frac{Z_1Z_2}{Z_{o\pi}}$

$Z_1 = 2\ [\ Z_{OC} - \sqrt{Z_{oc}(Z_{oc} - Z_{sc})}]$

$Z_2 = \sqrt{Z(Z - Z)}$

π – Network

$Z_{in} = Z_{out/uit}$

$Z_{\pi T} = \sqrt{Z_{oc}Z_{sc}}$

$Z_{OT} = \frac{Z_1Z_2}{\sqrt{\frac{Z_1^2}{4} + Z_1Z_2}} = \frac{Z_1Z_2}{Z_{OT}}$

$Z_1 = \frac{Z_{sc}[Z_{oc} \pm \sqrt{Z_{oc}(Z_{oc} - Z_{sc})}]}{Z_{oc} + \sqrt{Z_{oc}(Z_{oc} - Z_{sc})} - Z_{sc}}$

$Z_2 = \frac{1}{2}[Z_{oc} \pm \sqrt{Z_{oc}(Z_{oc} - Z_{sc})}\ ]$

ASYMMETRICAL FOUR – TERMINAL NETWORK

L-Section Image Impendance

$Zo_1 = \sqrt{Z_1^2 + Z_1Z_2}$

$Zo_2 = \frac{Z_1Z_2}{\sqrt{Z_1^2 + Z_1Z_2}}$

L-Section Iterative Impendance

$Zo_1 = \sqrt{\frac{Z_1^2}{4} + Z_1Z_2} + \frac{Z_1}{2}$

$Zo_2 = \sqrt{\frac{Z_1^2}{4} + Z_1Z_2} - \frac{Z_1}{2}$

Network Conversions

T →π Conversion

$Z_A = \frac{Z_1Z_2 + Z_2Z_3 2 + Z_1Z_3}{Z_2}$

$Z_B = \frac{Z_1Z_2 + Z_2Z_3 2 + Z_1Z_3}{Z_3}$

$Z_C = \frac{Z_1Z_2 + Z_2Z_3 2 + Z_1Z_3}{Z_1}$

π→T Conversion

$Z_1 = \frac{Z_AZ_B}{Z_A + Z_B + Z_C}$

$Z_2 = \frac{Z_BZ_C}{Z_A + Z_B + Z_C}$

$Z_3 = \frac{Z_AZ_C}{Z_A + Z_B + Z_C}$

Attenuators In General

dB = 8.686 X Neper

$dB = 10 \log(\frac{P_{in}}{P_{uit/out}})$

$dB = 20 \log N$

Insertion Loss = $10\log\frac{P_1}{P_2} = 20\log\frac{I_1}{I_2}$

Symmetrical Attenuators

$N = \frac{I_{in}}{I_{out/uit}} = \sqrt{\frac{P_{in}}{P_{out/uit}}}$

T-Type

$R_a = \frac{R(N-1)}{N+1}$

$R_b = \frac{2RN}{N^2+1}$

Π –Type

$R_a = \frac{R(N-1)}{N-1}$

$R_b = \frac{R(N-1)}{2N}$

Bridged T

$R_a = R = Z_o$

$R_b = R(N - 1)$

$R_c = \frac{R}{N-1}$

Asymmetrical Attenuators

$N = \sqrt{\frac{P_{in}}{P_{out/uit}}}$

L-Type

$R_a = \sqrt{R_1(R_1 - R_2)}$

$R_b = \sqrt{\frac{R_1R_2^2}{R_1 - R_2}}$

MUTUAL INDUCTANCE

$M = \sqrt{L_p \times L_s} \times k$

$K = \frac{M}{\sqrt{L_p P_s}}$

$K_c = \frac{1}{\sqrt{Q_p Q_s}}$

$Z_c = \frac{(\omega M)}{Z_s}$

$f_1 = \frac{f_o}{\sqrt{1+k}}$

$f_2 = \frac{f_o}{\sqrt{1-k}}$

$Q_p = \frac{X_{LP}}{R_p}$

$Q_s = \frac{X_{LS}}{R_s}$

Dot rule

$L_T = L_1 + L_2 + 2M$

$L_T = L_1 + L_2 - 2M$

AM

$m = \frac{V_m}{V_C}$

$P_{DSB} = \frac{m^2 P_C}{2}$

$P_T = P_c\left(1 + \frac{m^2}{2}\right)$

$m_T = \sqrt{m_1^2 + m_2^2}$

$V_c = V_c \sin \omega_c t + \frac{m V_C}{2} \cos(\omega_c - \omega_m)t - \frac{m V_C}{2} \cos(\omega_c + \omega_m)t$

FM

$V_C = V_C \sin(\omega_{cos} t - m cos \omega_{Mt})$

$BW = f_{mod} \times$ (number of sidebands) $\times 2$

SHANNON-HARTLEY

$C = \Delta f \log_2\left(1 + \frac{S}{N}\right)$

$dB = 10 \log\left(\frac{\text{signal power}}{\text{noise power}}\right)$

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